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The Surface-Constrained Level Set Method for Multiphase Interfaces

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The level set method is one of the most important tools of the applied mathematician. It allows one to mathematically express the evolution of shapes and is used throughout the sciences to model phenomena ranging from the free-surface motion of fluids, to the processing of digital images. Nevertheless, the level set method has largely been limited to modeling the evolution of shapes in euclidean spaces. We extend the level set method to the multiphase surface-constrained setting where surfaces are defined by *point clouds*.

When describing the shape of interfaces on surfaces, the multiphase setting requires a new type of level set function. For this purpose, we have developed a generalization of the *signed distance function*, which we call the *signed distance vector field* (SDVF). Defined on the surface, the SDVF is a vector-valued function that encodes the multiphase information in the orientation of its vectors. Upon choosing a level set equation and setting the SDVF as its initial condition, the evolution of the multiphase geometries can be prescribed. We will show how the SDVF can be combined with *minimizing movements* (MM) to naturally accommodate constraints on the motion of each phase.

We will illustrate the application of our method to multiphase interfacial motion by solving the level set equations via surface-constrained MM. Here, the MBO and hyperbolic MBO algorithms (3), which approximate multiphase curvature flows, will be used as test problems. This enables the simulation of surface-constrained, area-preserving, multiphase interface evolution—currently only possible with our approach.

Keywords: Surface PDE, level set, interfaces, approximation methods, numerical analysis

MSC2020: 35R35

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